

Circulation Sports: A Category That Already Had Members and No Name

The Three Properties, the Conversion Operator, the Drill/Sport Differentia, and a Census of the Warm-Ups That Were Already Doing It

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Note. This paper is a **study**: it discloses a classification of sports, a procedure for converting one sport into another, a boundary, a census of published warm-up formats and a set of pre-registered predictions. It discloses no mechanism and nothing patentable; its findings are numbered, each with what would break it, in §13. It was first published on 2026-08-07, early, for a reason of its own: a category is defined by whoever specifies it first, and a category specified badly by an incumbent is harder to repair than one specified openly. Its sibling *The Sport That Says Your Name* (2026-07-23) discloses the **caller mechanism** of the sport *sey*; the present paper names the **class of sports** that mechanism belongs to, and claims nothing that belongs to the members it did not create. The title's *no name* means no shared name for the class in the sources this paper cites; the nearest named class, cooperative games, is drawn on a different line (§10). The category name is dedicated to the public domain and will never be registered as a mark — the same split already applied to *sey* (free) and B-SeyTM (the mark). **One pre-registration in this paper has already been run, and it returned a partial null against the authors' own prediction; §8 reports the failure before it reports what survives.**

Abstract

This study classifies **cooperative and non-competitive ball games, keep-up circle games and sports warm-up drills** by their **scoring unit**. Some sports score the moment an exchange **ends** — the ball goes in, the rally dies, the opponent fails. Some score the moment an exchange **continues**. The second kind has been played for more than a thousand years and has members across East and Southeast Asia and in the warm-ups of mainstream sports. This paper names the class, specifies it, and draws its line against the nearest named class, cooperative games (§10).

A **circulation sport** is defined here by three properties held together: it scores the **completed transfer** rather than the termination; it **reaches every participant**, as a rule that makes skipping someone impossible rather than merely impolite; and it **accumulates to no one**, keeping no individual record. The three are not a feature list drawn from one game. They are a transformation rule, and the paper states it as a three-step **conversion operator** that can be applied by someone else to a sport we have never played — with worked conversions of basketball, volleyball, soccer, and tennis, and with the boundary drawn as carefully as the interior (Ultimate Frisbee is out; the rondo fails on reach; golf is the antipode). Of the three, one carries a precondition that is not a choice at all: below three participants a completed transfer is indistinguishable from a return, so **three is the smallest number at which "forward" is defined** — a result belonging to Lévi-Strauss, to the kula literature, and to the evolutionary work on upstream reciprocity, which §2.5 cites rather than claims.

Three findings drove the paper and none of them flatter it. **First**, the operator is not new, because it has already been run — in the opposite direction, in public, with a date and a name attached. The Malay circle game *sepak raga bulatan*, in which players keep a rattan ball aloft together, was converted into a competitive net sport — in its best-documented run by Hamid Maidin in Penang in February 1945, with earlier net forms of the rattan-ball game recorded in Siam and at Seremban — and standardized as *sepak takraw* at a regional meeting in Kuala Lumpur in 1960. What this paper specifies is the **inverse of a documented historical operation**, and it claims no more than that. **Second**, applying the operator to any classic sport tends to produce something that already exists as that sport's own warm-up — pepper, mini-tennis, the rondo, passing lines — which makes the category an empirical question with a natural experiment already running wherever youth sport is coached, rather than a proposal. **Third**, the authors pre-registered a census of those warm-ups (P-CS5) predicting that reach guarantees would prove **rare**, and then ran it. **They are not rare**. Reach guarantees are common, in at least four distinct architectures. The pre-registered claim is falsified and is reported as falsified. What survives is narrower, was arrived at after the data, and is labeled accordingly: every reach guarantee in the census **purchases reach with predictability** — a queue, a fixed role rotation, or a memorized pattern all tell you when your turn is coming — and reach whose order stays unpredictable across unbounded play remains unattested in that census. That gap, not reach itself, is where the sibling paper's caller mechanism sits.

The paper also states the differentia that decides whether any of this describes **sports** at all: *a sport owns its own ending; a drill is stopped from outside*. Within the genus of physical skilled activity that separation is necessary though not sufficient, and it supplies the category's own life-or-death test — if converted forms are only ever played instrumentally, as warm-ups and means to the competitive game, then this is a taxonomy of drills wearing a taxonomy of sports. **Six predictions are pre-registered with instruments and nulls, one of them already resolved against us**, together with an annual review date, and the whole set is stated against the base rate it must beat: cooperative sport has been tried as an organized movement, had a real decade, and did not displace competitive sport. One distinction is offered as the reason this is not that movement returning: **a cooperative game removes the opponent; a circulation sport keeps the opponent and changes what counts as a point**.

Keywords: sport classification, game design, rule modification, cooperative games, non-competitive games, cooperative scoring, keep-up games, circle games, passing games, warm-up drills, youth sport, physical education, equal participation, turn-taking, receiver selection, generalized exchange, indirect reciprocity,

kemari, chinlone, sepak raga, sepak takraw, rondo, pre-registration, circulation sport, conversion operator, drill versus sport.

Terms

Names used in this paper and the standard terms an examiner or reader would search for them.

Term used here	Standard term
circulation sport	sport or game scored on completed passes rather than on goals or won rallies, with guaranteed equal participation and no individual statistics; a keep-up or passing game with a collective score
completed transfer (the scoring unit)	completed pass; consecutive-pass count; rally count; cooperative scoring
termination	conventional scoring by ending the exchange: goal, basket, rally won, opponent's error
reach; reach guarantee	guaranteed equal participation; mandatory rotation; all-must-touch rule; turn order
accumulates to no one	no individual statistics; team or group record only
conversion operator	rule-modification procedure that turns a competitive sport into a pass-scored, equal-participation format
reverse conversion	historical change of a circle game into a competitive net sport (sepak raga to sepak takraw)
warm-up census	coded desk survey of published coaching drills and warm-ups against stated criteria
reach architecture (queue, role rotation, round completion, turn-taking norm)	methods of distributing turns in a group drill
the predictability price	trade-off between guaranteed turn distribution and an unpredictable turn order
drill/sport differentia	whether an activity's ending is supplied by its own rules or by someone outside it (stopping rule)
minimum arity of three	smallest group in which indirect (generalized) exchange differs from direct reciprocity
caller mechanism (sibling paper)	external announcer of the next receiver; randomized turn assignment

Prior-Art and Non-Assertion Statement

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attempt to enclose them. No patent has been or will be sought on anything disclosed in this paper, and no trademark or other exclusive right in the category it defines or in that category's name, by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control, in any jurisdiction, at any time. **The authors and those entities commit not to assert any such right against anyone practising or teaching anything disclosed here, or using the term *circulation sport*.** The commitment is stated rather than implied, and is not conditioned on reciprocity, attribution, or field of use. Nothing here is a mechanism; what the paper discloses is the numbered list in §13, *Findings disclosed*. **The term "circulation sport" is defined here as the name of the class, and freed here, deliberately and permanently;** the phrase first appears in this corpus in the sibling paper's freed term *AI-called circulation sport* (2026-07-23). A category that one party owns is a category nobody adopts, and the value of this work — if it has any — is entirely in other people applying it to sports we will never play. The institution's own product mark (B-Sey™/SM) is unaffected and is not the subject of this paper; the split between a freed idea and a held mark is the same one already applied to the Khmer word *sey*, which is free, and B-Sey™, which is not.

Terms this paper defines and frees (as terms of this paper, not as a claim that no one has used the words before): **circulation sport, the conversion operator, the reach guarantee, the predictability price, the drill/sport differentia** as stated in §9, and **the warm-up census** as a survey method.

Terms inherited from this corpus's earlier publications are **cited, not re-claimed**: *equal-yet-random distribution, the caller mechanism, the cannot-target-weakness property, the dyadic scoring unit, start-line handicap, and the play/currency wall* all belong to *The Sport That Says Your Name* (2026-07-23) and are dedicated there.

Most of this paper's subject matter is not ours and is not claimed. Kemari, chinlone, sepak raga, jianzi, footbag, and *sey* itself are living or historical traditions belonging to the communities that carry them. Terry Orlick's cooperative-games canon and the New Games Movement are prior art of the most direct kind, and are treated in §10 as the position this paper has to answer rather than as background. Every warm-up drill surveyed in §8 was invented by coaches whose names are mostly unrecorded, and at least one of the operator's outputs is already published by a sport's governing body: USA Volleyball's *Simplified Youth Rules* describe "cooperative scoring" that counts each three-contact crossing of the net as a point for both teams together (§5.1, §10). General taxonomies of play — Roger Caillois's above all (*Les jeux et les hommes*, 1958; *English Man, Play and Games*, 1961) — sort games by the kind of play they offer (competition, chance, simulation, vertigo) and along a continuum from free play to rule-bound games; the classification here is narrower, by what a game scores, and is offered beside those taxonomies rather than against them. **The contribution claimed here is a definition, a procedure, a boundary, a falsification test, and one census — nothing more.**

The search behind this paper, with its aperture. One desk survey, P-CS5 (§8): 22 published warm-up, drill and icebreaker formats across basketball, volleyball, soccer, tennis, baseball, rugby and Ultimate, with the icebreaker and physical-education literature, run on 2026-08-07 — documented practice, not fieldwork. It falsified the authors' prediction that reach guarantees are rare, and left one survivor, the predictability price (§8.5): reach that is guaranteed and whose order stays unpredictable across unbounded play was not found in that aperture on that date. Two formats found on revision (2026-10-01), outside that census and not scored in it, bear on the survivor and on §10 and are reported where they bear (§8.5, §10). No literature

census has been run on the class itself, its name, or the drill/sport differentia, and nothing here asserts that they are new.

1 · Why an Institution Building a Gratitude Ledger Is Writing About Sport

This paper belongs to a body of work on **reciprocity infrastructure** — systems in which value moves because someone gave, rather than because someone paid. That work has a physical problem it cannot solve on a screen. Giving, at the scale where it changes a person, is learned in bodies and in groups, early, and mostly through play. A child who is never passed to has not learned a fact about passing; they have learned a fact about themselves.

The institution's operative unit is a **gift given forward**: something received and passed on, rather than received and returned. Sport is where that unit either exists in a person's motor memory or does not. So the question this paper asks is narrow and practical. *Which sports are built out of the give-forward unit, and is there a procedure for building more of them?*

It turns out that some are, that they are old, that they are scattered across several cultures — some related by descent, as kemari is to the Chinese ball game it came from, and some not — and that they are usually described one game at a time rather than as a class. That is the whole occasion for this paper. **We are not proposing that competitive sport is a mistake.** Competition is one of the great engines of skill, and this paper's central distinction (§10) exists precisely to keep the opponent rather than remove them. What is proposed is that a second family of sports exists alongside the competitive one, that it has been under-specified as a family, and that its members would be easier to make on purpose if somebody wrote down how.

A note on tone that the rest of the paper tries to keep. It is easy to write about non-competitive play by cataloguing what competitive play does to children. That register is available and this paper declines it. **The claim here is about what circulation is, never about what competition lacks** — partly because the second claim is not one the evidence supports, and partly because a taxonomy written as an indictment gets read as an indictment and not as a taxonomy.

2 · The Category, Specified

A category needs a transformation rule rather than a list of one game's features, or it is a description of that game with ambitions. Three properties, held **together**:

2.1 It scores the completed transfer, not the termination

In classic competitive sport the scoring event is an **ending**. The ball goes through the hoop; the rally dies on your opponent's side; the puck crosses the line. Value is created by *ending the exchange in your favor*. The transfer — the pass, the assist, the set, the feed — is instrumental to that ending and is recorded, if at all, as a supporting statistic.

In a circulation sport the polarity is inverted. **Value is created by the exchange continuing.** The scoring unit is a transfer that arrived: a pass completed, a rally sustained, a ball kept up. The termination still exists and is still necessary — a rally must eventually end — but it is demoted to a non-scoring event, and in the sharper members it is the thing that *stops* the count rather than the thing that increments it.

2.2 It reaches everyone

This property is not a fairness guard bolted onto a game. It is what the word *circulation* means. A circulatory system that skips an organ is not a circulatory system operating imperfectly; it is a pathology with a name. Blood that does not reach the extremities is the definition of circulatory failure.

So the requirement is strict, and the strictness is the point: **skipping a participant must be impossible as a move, not merely discouraged as a habit.** A rule that says "try to include everyone" does not satisfy this. A bonus for including everyone does not satisfy it either — §8 finds exactly such a bonus in the wild and classifies it as a near-miss. What satisfies it is a distribution rule that removes the choice: a mandatory rotation, a called receiver, a round that is not complete until every participant has received.

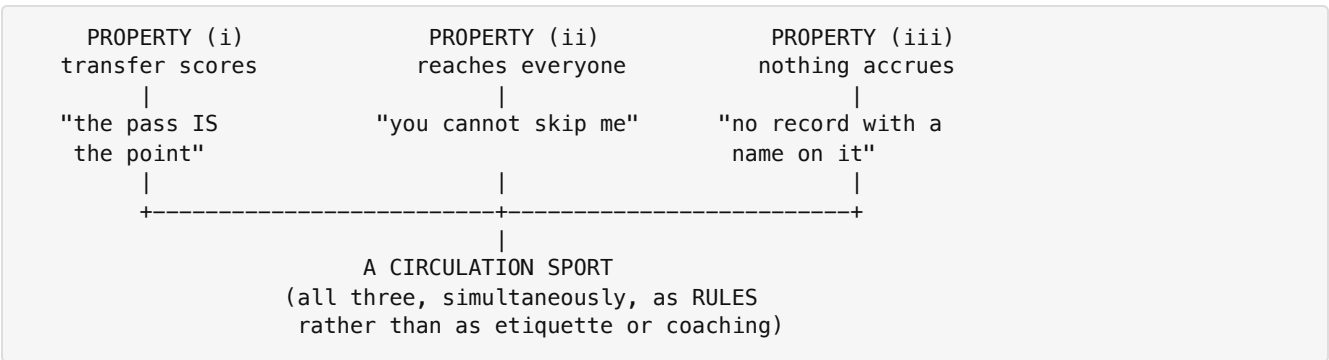
2.3 It accumulates to no one

Classic sports are, among other things, **archives of individual accumulation.** Career goals, batting averages, personal bests, the ranked ladder, the hall of fame. The archive is not incidental to those sports; for many participants it is the point, and it is a large part of what makes elite competition legible and worth watching.

A circulation sport refuses the archive. No player statistics, ever. Counts that run down rather than up, or streaks that always eventually die, or records that belong to the circle rather than to any person in it. The test is simple: **after the session, is there any number attached to an individual that persists?** If yes, the property fails.

2.4 The three together, and why the conjunction matters

Each property alone is common. Together they are not.



The conjunction is what does the work, and §6 and §8 are both exercises in showing how many familiar activities hold two of the three.

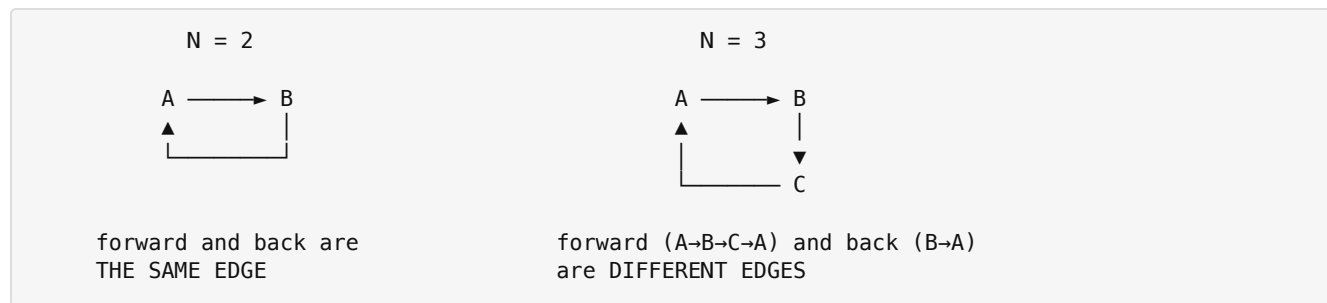
2.5 The minimum arity: why the circle is three

Added 2026-08-22. The three properties above are stated as design choices, and two of them are. This section reports the one that is not — and it is a result the paper stands on rather than announces, because it belongs to anthropology and to evolutionary dynamics and has for decades.

The question arrived from the wrong end. Asked whether the sibling paper's game could be played by two people as a simple name courtesy — one says the other's name, the other says it back — we answered no, and then had to work out why, because the honest first answer was only that it did not feel like the game. Two people can pass a ball, sustain a rally, take turns, and enjoy an hour of it. What two people cannot do is **circulate**.

At $N = 2$, every pass forward is a pass back. The second player has exactly one address to send the thing to, and it is the address it came from. Forward and back are the same edge. Which means the unit this category is built on — a transfer that continues *away* from its source — has no way to distinguish itself from the unit it is defined against, a transfer that returns. What a dyad has is reciprocation, and reciprocation is not a defective circulation; it is a different and entirely respectable thing, with its own long literature and its own social work to do.

At $N = 3$ the two motions come apart for the first time. A passes to B, and B's forward option (C) is now a different edge from B's return option (A). The choice exists, and so the category does.



Three is therefore the smallest number at which "forward" is defined. Not the smallest number at which the game is fun, or fair, or worth organizing — those are different questions with different answers. It is the smallest number at which the category's own unit is distinguishable from its complement, which makes it a fact about the definition in §2, not a fact about anybody's product.

The result is not ours, and it has been settled three times in three literatures. In kinship theory, Lévi-Strauss's *Les structures élémentaires de la parenté* (1949; English: *The Elementary Structures of Kinship*, Beacon Press, 1969) draws exactly this line, calling the two-group case **restricted exchange** — direct and immediate, A and B trading with each other — and the three-or-more case **generalized exchange**, in which A gives to B, B to C, and C back to A, delayed and indirect, and capable of integrating an indefinite number of groups precisely because it is not a swap. In economic anthropology the attested instance is the **kula** ring of the Massim islands, which Malinowski studied from the Trobriands (*Argonauts of the Western Pacific*, 1922): a circuit of many islands around which two classes of valuables travel in opposite directions — shell necklaces one way, armshells the other — so that each object moves on around the ring rather than back to the partner it came from. And in evolutionary dynamics the same threshold follows from the formal definitions: **upstream reciprocity** — help someone because someone helped you — is distinguished from

direct reciprocity by exactly this arithmetic, since the shortest possible cycle of indirect reciprocity is three players, a two-player cycle being direct reciprocity by definition (Boyd & Richerson, "The evolution of indirect reciprocity," *Social Networks* 11(3), 1989; Nowak & Roch, "Upstream reciprocity and the evolution of gratitude," *Proc. R. Soc. B* 274(1610): 605–610, 2007).

That last paper's title should be read as the caution it is. The mechanism this institution builds on has been modelled, named, and published under the word *gratitude* for nearly twenty years. **We are late to it, and the correct posture is grounding, not discovery** — the same posture §8 arrived at the hard way when the census returned a null against our own prediction.

One conflation to avoid, because we nearly committed it. Sahlins's celebrated triad — **generalized, balanced, and negative** reciprocity ("On the Sociology of Primitive Exchange," in Banton, ed., *The Relevance of Models for Social Anthropology*, ASA Monographs 1, Tavistock, 1965; reprinted in *Stone Age Economics*, 1972) — sorts exchange by **social distance and the expectation of return**, not by the number of parties. Sahlins's *generalized reciprocity* means open-ended giving among close kin with no reckoning of equivalence; Lévi-Strauss's *generalized exchange* means a cycle of three or more. **The two share a word and not a claim**, and a paper that cited them as one argument would be citing prior art wrongly, which is worse than not citing it at all.

What this does for the definition in §2. It splits the three properties by their modal status, which they did not previously have. Properties (ii) *reaches everyone* and (iii) *accumulates to no one* remain design choices: a designer could build a transfer-scoring sport without either, and §6 and §8 are full of activities that do. Property (i) — scoring the completed transfer — is a design choice too, but it carries a **precondition** that is not: below three participants there is no completed transfer distinguishable from a return, and the property cannot be satisfied because it cannot be stated. The definition thus rests on one forced floor and two chosen walls, and saying so makes the category slightly smaller and considerably more honest than it was when all three properties looked alike.

What is deliberately not claimed here. Not that three is a good size — it is a floor and a poor ceiling, and most attested members of the category run much larger. Not that the floor implies any particular quorum for any particular product, which is a design question this paper does not touch. And not, emphatically, that the arithmetic makes a dyad's kindnesses lesser: a couple passing a thing back and forth is doing something the human record values enormously, and the only claim made is that it is not the thing this paper is naming.

3 • The Members That Already Existed

The single most important sentence in this paper is that **we did not invent this category, and its best members predate us by centuries**. A reader who knows the field and finds these uncited would be right to stop reading.

Kemari (蹴鞠, Japan). A court football that came to Japan from China, derived from the Chinese ball game *cuju*, and was popular in the Heian (794–1185) and Kamakura (1185–1333) periods. Its first date is uncertain: a ball game recorded in the *Nihon Shoki* at Hōkōji in 644 is often cited as the earliest reference, but that game may not have been kemari, and the earliest reliable record of kemari by name, written in the

mid-Heian period, places games in 701. Eight players stand in a circle and kick the ball to one another so that it stays in the air, passing it with control and precision. It is non-competitive: the goal is to keep the ball from touching the ground, together, for as long as possible. Its rules were codified from the early thirteenth century, when the art of *kemari* took shape under the shogun Minamoto no Yoriie. Fearing its loss to Japan's rapid modernization, Emperor Meiji supported the association founded in 1903 to preserve it with an imperial grant. **Kemari satisfies all three properties — its reach held by the convention of the circle, which §8.4 discusses — and has been played for more than a thousand years.**

Chinlone (ချင်းလုံး, Myanmar). Widely described as Myanmar's traditional national sport, typically played by six people as one team with a hand-woven rattan ball. In its traditional form there is no opposing team and no winning or losing. Success is measured by how long the ball stays airborne and how gracefully each touch is executed; one player occupies the center of the circle. It is commonly described as closer to a performance or a dance than to a contest. (It too has been moved toward competition: a rulebook written in 1953 at the Burmese government's order made it more competitive, and it was a contested event at the 2013 Southeast Asian Games.)

Sepak raga bulatan (Malay Archipelago). The circle form, generally treated as the older one: five to ten players stand in a ring and keep a woven rattan ball aloft with feet, knees, chest and head, never the hands, the ball passing from player to player, with the emphasis on skill and endurance rather than on winning. Its documented history reaches the Malacca Sultanate of the fifteenth century: the *Sejarah Melayu* (Malay Annals) records a game in which the ball knocked a prince's headdress to the ground; he killed the player who had kicked it, a son of the chief minister, and the crisis that followed ended with the prince sent from Malacca to rule Pahang — which is to say the game was consequential enough to appear in a court chronicle.

Jianzi (毽子, China), **footbag circles**, and **keepie-uppie** are the same shape at smaller scale and with different objects.

Sey (សី, Cambodia) is the game this institution's own sport descends from — a folk game played at dusk, inherited rather than invented, in which a circle keeps an object alive together. Its treatment as a *design* rather than an inheritance would be the same error this section exists to prevent.

The lineage is worth stating plainly because of where it sits: **four of the strongest members of this category are Asian circle games, three of them Southeast Asian, and the specification is being written by an institution with a Cambodian-born founder, about a Khmer game.** That is a reason for care rather than for pride. The tradition is older than the theory and does not require it.

4 · The Conversion Has Already Been Run — In the Other Direction

This is the finding that most changed the paper: it leaves the paper no claim that the operation itself is new, and that makes the paper stronger rather than weaker.

The Malay circle game did not merely coexist with a competitive net sport. **It was converted into one, by a named person, on a date — and not only once.** In February 1945, in Penang, Hamid Maidin (also spelled Mydin) introduced a net (*jaring*) and rules that accounts liken to volleyball's or to badminton's, inviting

Mohamad Abdul Rahman — described in one history of the sport as the best player of the circle game — and Syed Yaacob to try it; the first organised competition followed that May, and the Malayan federation founded in 1960 recognised Maidin as the creator of the modern game. Earlier net play is also recorded: at Seremban in 1935, where sepak raga was played over a net on a badminton court, and in Siam, where the Siam Sports Association drafted competition rules in 1929 and introduced a volleyball-style net four years later. In 1960 delegates met in Kuala Lumpur, standardized the rules, and adopted the name **sepak takraw**, under which the sport is now governed internationally; Singapore's National Heritage Board lists the delegations as Malaya, Singapore, Indonesia, Laos and Thailand, and other accounts include Myanmar.

Read against §2, the 1945 run is a documented, attributed run of the conversion operator **in reverse**:

sepak raga bulatan		sepak takraw
-----		-----
circle, passed around	Feb 1945	two sides, a net
keep it aloft together	-----→	end the rally on their side
no score, no sides	(jaring)	points, sets, matches
reaches everyone	1960	targeting the weak side is
accumulates to no one	(naming)	a legitimate tactic
		individual and team records
THREE PROPERTIES HELD		THREE PROPERTIES RELEASED

Three consequences follow, and all three are load-bearing.

The operator is real. It is not a theoretical construct invented to justify one product. A human being ran it in one direction in 1945, deliberately, with a design intent that is legible eighty years later, and others had run it before him. A procedure that has been executed has a much stronger claim to being a procedure than one that has only been described.

The direction is a choice, not a gradient. It is tempting to narrate sport as evolving from folk play toward formal competition, with circulation as the primitive stage. The 1945 conversion, and the earlier ones in Siam and at Seremban, are better read as *options that were taken*: the circle form did not die of natural causes, and in fact it did not die at all — chinlone and kemari are both alive, and the *bulatan* form persists. Somebody added a net, more than once, and organised competition followed quickly — in Penang within three months. The inverse operation is equally available. Its nearest written form is the cooperative-games literature, from which §10 distinguishes the form §5 writes down.

Our contribution shrinks to the right size. This paper does not propose a new kind of sport. It specifies the inverse of an operation that has been performed, names the class the pre-conversion games belong to, and asks whether the inverse can be run deliberately. That is a smaller claim than "a new category of sport," and it is the one the evidence supports.

Sources for §3–§4, checked on revision (2026-10-01): the Imperial Household Agency's account of kemari; the English Wikipedia articles Kemari, Chinlone, Sepak raga and Sepak takraw (History), and the sources they cite; the Sepak Takraw Association of Canada, International Sepak Takraw History (the 1945 account and the description of Mohamad Abdul Rahman); Singapore's National Heritage Board, Roots.gov.sg, Sepak Takraw (the 1960 delegations). Where these accounts differ, §4 says so.

5 • The Conversion Operator

A named operator that is never written is a promise, not a contribution. Here it is as a procedure, stated so that someone who has never spoken to us can apply it to a sport we have never played.

- (i) **RE-SEAT THE SCORING UNIT.** Find the sport's existing *transfer* event — pass, assist, set, feed, handoff, rally stroke — and make that the point. Demote the termination to a non-scoring necessity.
- (ii) **GUARANTEE REACH.** Add a distribution rule that makes skipping or targeting a participant impossible *as a move*: a called receiver, a mandatory rotation, an all-must-touch requirement, or a round that does not close until everyone has received.
- (iii) **DELETE ACCUMULATION.** No individual statistics, ever. Records belong to the unit. Counts run down, or die, or are not kept.

5.1 Worked conversions

The operator must produce **playable games** or the category is one sport wearing a taxonomy. Four worked examples, chosen because they span net, invasion, and racket sports:

Sport	(i) transfer as the point	(ii) reach	(iii) no accumulation
Basketball	the assist scores; a basket without a preceding pass is worth nothing	all five must touch before a shot may be taken	team streak only; no box score
Volleyball	completed three-touch sequences across the net , counted by both sides together	rotation already exists; extend it to guaranteed contacts	the rally that never dies is the record
Soccer	possession-chain length; a goal counts only after N unique touchers	the N-unique-touchers requirement is the reach rule	chain records, never scorers
Tennis	rally count	strict alternation is inherent to the sport	the rally dies and nothing carries

An honest note about these outputs, which §7 will develop: **none of them are inventions**. Each is close to an existing warm-up, youth rule or coaching game; the volleyball row's first and third columns closely match the "cooperative scoring" option in USA Volleyball's *Simplified Youth Rules*, in which "the ball is the opponent" and both teams count each three-contact crossing of the net as a point. That is not an embarrassment for the operator; it is the strongest available evidence that the operator describes something real. *The operator's value is that it explains what these have in common, not that it produces novelties.*

5.2 The scope guard

The operator is a **corpus contribution and not a product roadmap**. This institution's own sport is deliberately scoped to the first fifteen minutes of a practice and is explicitly not a platform for all sports. Publishing a general conversion procedure while keeping the product narrow is the same split applied elsewhere in this corpus: **keep the product narrow and the idea broad**. Other people convert their own sports; that is the intended outcome, and the CC0 dedication exists to make it frictionless.

6 • The Negative Space

A category that admits everything explains nothing. The boundary is drawn here with more care than the interior, because the boundary is where the definition is tested.

Ultimate Frisbee — OUT, and it is the cleanest near-miss. Ultimate's Spirit of the Game is a genuine and admirable normative achievement, and it is *governance*: it regulates conduct, self-officiating, and dispute resolution. It does not touch the scoring unit. A point in Ultimate is scored by catching the disc in the end zone — a termination, though the catch is itself a completed pass: it counts here as a termination because it ends the point and is the only scoring event — and defense consists of preventing that. Ultimate is a well-governed competitive sport, which is a fine thing to be. **Its exclusion is what proves the category is about the scoring unit rather than about niceness**, and any definition that lets Ultimate in has quietly become a definition of good sportsmanship.

Relay races — PARTIAL. The transfer is real and ceremonial (the baton exchange is the most transfer-centric moment in track), and reach is guaranteed by the structure. But the scoring event is a finish line, and results accumulate to teams and to individuals in the record book. Two of three.

Rondo / piggy-in-the-middle — FAILS (ii), and instructively. The rondo scores consecutive completed passes, which is property (i) done well and is precisely why it is such a beloved warm-up. But it has a **designated loser** in the middle, and the defender who wins the ball changes places with the player who lost it. Targeting the weakest passer is not a violation of the rondo; it is how you get out of the middle. *Reach is not merely unguaranteed here — its opposite is the objective.*

Shooting lines and batting practice — FAIL (i) and (iii). There is no transfer at all, and makes are counted individually. Pure individual accumulation in a warm-up costume.

Cooperative board games — OUT. No circulating physical token and no bodies; whatever they are, they are not sports, and stretching the category to reach them would cost the definition its shape.

Golf — THE ANTIPODE. No transfer, complete individual accumulation, and a scoring unit that is a termination repeated. Golf is useful precisely as the category's polar opposite — and note the one circulatory organ it does contain: **the handicap**, a mechanism whose whole purpose is to let unequal players share a game, which is the reach property arriving through the back door of a sport built without it.

7 • The Discovery: Classic Sports Already Contain Their Circulatory Forms

Applying the operator turned up something better than the operator.

Its outputs already exist, scattered, as warm-ups and youth rules. Pepper. Mini-tennis rallies. Passing lines. The rondo. Five-passes-before-you-shoot. Keep-it-up. The butterfly drill. They are catalogued sport by sport, as each sport's own warm-ups; this paper reads them as one family — the volleyball coach and the soccer coach and the tennis coach have each built the circulatory form of their own sport, called it a warm-up, and used it for fifteen minutes before playing the real one.

This matters more than the taxonomy does, because it converts the category from a proposal into an **empirical question with a natural experiment already running wherever youth sport is coached**. We do not have to argue that circulation sports would be enjoyable. We can go and look at whether the ones that already exist are.

It also promotes what had been the weakest of the founder's four predictions into the strongest. "*Warming up with sey gets extended longer and longer*" had been filed as a test of whether a drill is fun. If warm-ups **are** the circulatory forms of their own sports, then a warm-up that keeps getting extended is **the circulatory form eating the competitive one, measured inside a single session** — the category thesis in miniature, at the smallest scale at which it could possibly be observed.

8 · The Census (P-CS5), Run — And the Null That Fired

8.1 What was pre-registered, and when

On **2026-08-05**, before any survey was conducted, the authors recorded the following prediction in their dated project record (a private, version-controlled archive; this paper, first published on 2026-08-07, is the prediction's first publication):

P-CS5 — the warm-up census. Survey what mainstream sports do in warm-ups and score each format against the three properties. *Predicted shape: properties (i) transfer-as-the-point and (iii) no-accumulation are common; property (ii) reach is rare — and (ii) is where the sibling paper's entire contribution sits.* **Stated null: if reach guarantees turn out to be common, the uniqueness claim weakens materially.**

The census was run on **2026-08-07** as a desk survey of published coaching materials across basketball, volleyball, soccer, tennis, baseball, rugby, and Ultimate, together with the icebreaker and PE literature. It is a survey of **what is documented**, not an observational study of what coaches actually do, which is a limitation §12 states rather than minimizes.

8.2 Scoring convention

One methodological decision has to be declared because it changes the answer. A drill with **no score at all** cannot satisfy property (i): there is a difference between a scoring unit seated on the transfer and no scoring unit at all. The census therefore scores three values rather than two:

- ✓ — the property is actively instantiated by a rule
- ○ — vacuous: the property is not violated, but only because the relevant machinery is absent
- ✗ — violated

Second: property (ii) at **n = 2** is trivially satisfied, because a two-person drill reaches everyone by arithmetic rather than by design. Two-person formats are marked and are not counted as evidence for reach.

8.3 The census

Format	Sport	(i) transfer	(ii) reach	(iii) no accrual	Note
Kemari	—	✓	✓	✓	full member; rules codified from the 13th c.
Chinlone	—	✓	✓	✓	full member; no winning or losing (traditional form)
Sepak raga bulatan	—	✓	✓	✓	full member; the ball passes player to player
Cooperative mini-tennis rally	tennis	✓	✓ (n=2)	✓	target 10–15 consecutive balls
Pepper (2-person)	volleyball	✓	✓ (n=2)	○	control count; dies on a drop
Rondo (4v2)	soccer	✓	✗	✓	designated loser in the middle
Completed Passes Game	basketball	✓	~	○	reach <i>bonus</i> , not reach rule
Ten-in-a-row passing	volleyball	✓	○	✓	count dies at a drop
Butterfly drill	volleyball	○	✓	○	fixed role rotation
Three-man weave	basketball	✗	✓	○	ends in a layup; queue rotation
Line-pass warm-up	rugby	○	✓	○	"until every player has run front to back"
Three-person marking	Ultimate	○	✓	○	mark → throw → go, roles rotate
Name Toss	icebreaker	○	✓✓	✓	round closes when all have received
Passing Name Game	soccer	○	✗	○	the <i>passer</i> names their own choice
Pepper (batter + fielders)	baseball	○	✗	○	fixed hub; a star, not a circle
Five-passes-before-you-shoot	basketball	~	✗	○	passes are a precondition, not the point
Keep-away / possession grids	soccer	✓	✗	✓	free choice of receiver
Queen of the Court	volleyball	✗	✗	✗	scores side-outs; the monarch stays
Shooting lines	basketball	✗	✓	✗	queue reach; individual makes counted
Batting practice	baseball	✗	✓	✗	queue reach; individual accumulation
Ultimate (the sport)	Ultimate	✗	✗	✗	governance, not scoring unit
Golf	golf	✗	✗	✗	the antipode

8.4 The result, stated against the prediction

Property (i) — transfer as the point: CONFIRMED, with a qualification. Among formats that score at all, seating the score on the transfer is common and unremarkable: the rondo counts consecutive passes, cooperative tennis counts the rally, pepper counts control, the youth basketball game counts completed passes. The qualification is that **most warm-ups do not score at all**, so the property is more often *vacuous* than *instantiated*. The prediction said "common" and meant instantiated; the honest reading is "common where scoring exists, and scoring often does not."

Property (iii) — no accumulation: CONFIRMED, and largely for an uninteresting reason. Warm-ups mostly keep no records because they are warm-ups, not because anyone decided that records were the problem. Where individual counting does appear — shooting lines, batting practice, Queen of the Court — it appears immediately and without apology.

Property (ii) — reach: FALSIFIED. The prediction was that reach guarantees would be rare. **They are not rare.** The census found them in at least four distinct architectures:

A. THE QUEUE	three-man weave, rugby line, shooting lines, batting practice → reach by waiting your turn in a line
B. FIXED ROLE ROTATION	butterfly drill, Ultimate 3-person marking → reach by cycling through positions
C. ROUND COMPLETION	Name Toss → reach by "everyone receives exactly once"
D. TURN-TAKING NORM	kemari, chinlone, sepak raga bulatan → reach by an inherited convention of the circle

The stated null therefore fires: **the uniqueness claim weakens materially, and this paper reports that before it reports anything else.** The sibling paper's caller mechanism is not the first mechanism to guarantee that everybody touches the ball. It is not close to the first. Coaches have been guaranteeing reach with lines and rotations for as long as there have been lines and rotations.

Revision note (2026-10-01): architecture D and the full members. Architecture D is the weakest of the four as a guarantee in §2.2's sense. For the three traditional games the census recorded reach from secondary descriptions of players passing among themselves around a ring, and the sources checked on revision give no written rule that every player must receive. Their reach is held by the convention of the circle, which §2.2, read strictly, would class as a habit rather than a rule that removes the choice. The census scored them ✓ and the scores stand as recorded; a stricter reader would score their reach ~ and treat the three games as near-members, in which case no format in the census holds all three properties as rules. Whether an inherited convention of a ritualised circle counts as a rule in §2.2's sense is left open (§12). The falsification above does not depend on D: architectures A, B and C are rules. Until this revision the census note for *sepak raga bulatan* read *explicit turn-taking*; the sources checked on revision describe the ball passing from player to player, not a fixed order, and the note now says so.

8.5 What survives, explicitly labeled as post-data

The following was arrived at **after** seeing the census and is therefore not a pre-registration, not a confirmed hypothesis, and not entitled to any of the credit the falsified prediction forfeited. It is stated because it is what the data actually shows, and hiding it would be worse than labeling it.

The predictability price. Every reach guarantee found in the census purchases reach with **predictability**. A queue tells you exactly when your turn is coming. A role rotation tells you where you are going next. A memorized name pattern is unpredictable exactly once and then becomes a fixed order by design. The turn-taking norm of the old circle games is a social convention that is legible precisely because it is regular.

Reach whose order remains unpredictable across unbounded repeated play is unattested in this census.

This is not, however, a new idea in this corpus, and the census does not get to claim it as a discovery. *The Sport That Says Your Name* (2026-07-23) already argued the same point from the armchair, in the course of justifying its own selection policy: "*Equality without randomness would be a queue (predictable, boring, and stigmatizing — everyone can see whose turn is being "given"); randomness without equality is every pickup game ever played (the ball orbits the skilled). The conjunction is the mechanism.*" That sentence was written from design reasoning, before any survey existed. **What §8 adds is not the insight but its empirical status:** a claim that had been asserted about the space of possible designs is now a documented finding about the designs the census surveyed. The upgrade is from *plausible* to *attested*, which is worth something, and it is considerably less than a new finding would be worth.

Two observations follow, and neither is tested.

The gap is narrow and specific. What is unattested is not reach, and not unpredictability, and not name-calling — the Passing Name Game already puts names in a circle, and the Name Toss already closes a round on full coverage. What is unattested in the census is the **conjunction:** reach that is guaranteed, in an order nobody can anticipate, sustained indefinitely rather than for one round, inside an activity that also keeps a score and owns its own ending. Every component exists; the combination does not appear in the surveyed material. (The sibling paper has since scoped its own caller to the same limit: within a round dealt as a shuffled roster, the last name is forced for anyone counting, so its order is unpredictable for the circle rather than at every call — *The Sport That Says Your Name*, §4.4, 2026-09-23.)

Found on revision (2026-10-01), outside the census and not scored in it. A published physical-education game, *Speed Ball* (The Physical Educator), opens with a stage in which each team passes a ball from its own baseline to the opposite one, under the rule that "all players must touch the ball before reaching the opposite baseline", and leaves the order to the passers. Reach there is a rule, renewed on every run, and the order is not fixed in advance — but it is the passers' choice, known to the passer and open to favouring or targeting within each run. It therefore does not meet the gap as stated above (an order *nobody* can anticipate). Against P-CS6 as registered, which excludes only formats with an external announcing party (§11), it is a candidate for the null: whether it resolves the prediction depends on reading *unpredictable* as unpredictable to everyone or only to the receiver. That reading is left to the prediction's review, and P-CS6 is not reworded here.

There may be a reason the combination is missing, and it is not flattering to assume it is oversight. Guaranteeing reach in an order that no player chooses requires something to *hold* the order — an impartial party who knows whose turn is next and announces it. Before cheap devices that holder would usually have been a person outside the play, a coach or a non-playing caller (a drawn lot or a shuffled deck can hold an order, but someone still has to read it out), and a coach calling every single receiver for twenty minutes is a coach doing nothing else. That is a hypothesis about **why** the gap exists, it is untested, and it is exactly the kind of just-so story that a category-defining paper should flag rather than deploy.

There is also a possible mechanism connecting this to §9's falsification, offered as a conjecture with $n = 0$: **predictable reach and sustained attention may be in tension.** If you know your turn is four away, there is nothing to attend to until then, and the line looks at the ball rather than at each other. If that is right, it would explain why formats in architectures A and B have stayed warm-ups — not because they are badly designed, but because a queue is a very good way to distribute touches and a very poor way to hold a room. **We do not know this. It is the most interesting thing the census produced and the least supported.**

9 · Drill or Sport?

The category's own life-or-death test, and the definition it requires.

9.1 The falsification

If the converted forms are only ever played instrumentally — as warm-ups, PE units, means to the competitive game — and never voluntarily for their own sake, then this is not a category of SPORTS. It is a category of DRILLS.

The census makes this sharper rather than safer. Nearly every mainstream member found in §8 sits inside a practice session, before the real game, and stops when a coach says so. If that is the whole story, the honest conclusion is that circulation is something sports do to warm up, not something a sport can be — and that the three properties describe a preparation phase rather than a class.

9.2 The differentia

The test requires knowing what separates the two, and the answer that survives scrutiny is structural rather than attitudinal:

A sport owns its own ending. A drill is stopped from outside.

A drill's boundary is exogenous: the whistle, the coach, the clock on the wall, the start of the real practice. A sport's boundary is endogenous: the ball drops, the count reaches zero, the rules themselves say *that was the end*.

Register it correctly: **this is a differentia within a genus, not a definition of sport.** Within the genus of physical skilled activity it separates drill from sport. It does not distinguish sport from coin-flipping, and owning your ending is **necessary, not sufficient**. Claiming otherwise would be an overclaim this paper cannot carry.

Two rival criteria were considered and are better understood as **consequences** of this one:

- **Terminal vs. instrumental** (*is it played for its own sake?*) is an attitude, so the same activity is a drill for one person and a sport for another. True, unobservable, and unable to define anything.
- **Story vs. capability** (*what does it leave behind?*) is an output, and is the sharpest of the three: a sport is play that generates a story; a drill is play that generates a capability.

The chain that makes the structural criterion primary:

OWNS ITS ENDING (structure) → HAS A STORY (output) → IS WORTH RETURNING TO (memory) → IS TERMINAL (attitude)

A story requires an ending the activity itself supplied. "*And then the streak died at eighty-seven*" is a story. A drill that stops when the whistle blows has no ending, only an **interruption**, and there is nothing to tell. One structural fact produces the other two.

9.3 The staircase

Applied to the institution's own sport, the differentia produces an uncomfortable and clarifying result. Its three modes were designed in July for other reasons entirely, and they line up as a ladder across the drill/sport boundary:

Mode	Ending condition	Verdict
Friendship	none — it stops when people stop	a drill
Streak	ends when the object drops	a sport
Team	ends at zero	a sport

The floor of the ladder is a drill and this is the correct result, not an embarrassment: the no-score mode is the icebreaker, the first fifteen minutes, the thing you do so that everyone knows each other's names. **The graduation from the no-score mode to the counted one is the exact moment the activity acquires its own ending** — which is to say, becomes a sport.

Note also what this makes of the machinery around a collective streak: park records, the repeated-digit milestones, "*our circle's record is eighty-seven*." Read through the chain above, those are **story-manufacturing equipment bolted onto a warm-up**. An activity engineered to push itself across the drill/sport boundary is a reasonable description of what is being attempted.

9.4 The honest counterexample

"*Run the passing drill until someone drops it*" owns its ending and is still called a drill.

This is not a counterexample; it is the thesis in miniature and a confirming instance. The moment you run a drill to a natural ending **and start remembering the number**, you have converted it into a sport. That is precisely the founder's first prediction, restated as a boundary case — and it is the reason the definition makes the drill/sport line **thin and crossable**. Some would call thinness a weakness in a definition. Here it is the claim: a definition that made the line thick would contradict a category that exists at the crossing.

10 • The Base Rate: Circulation Is Not Cooperation

Every prediction in §11 is, in effect, a prediction that *this time it works*. A pre-registration that omits the base rate is naive, so the base rate goes first.

Cooperative sport has already been tried as an organized movement, had a real decade, and did not displace competitive sport.

Terry Orlick's *The Cooperative Sports & Games Book: Challenge Without Competition* (Pantheon, 1978) collected more than a hundred games built on cooperation rather than competition, for every age and setting; it was successful enough to be revised, republished, and followed by a second volume. The **New Games Movement**, emerging from the San Francisco Bay Area in the late 1960s around Stewart Brand, George Leonard, Pat Farrington, Bernie DeKoven and others, ran large public tournaments from 1973 to 1976 that drew thousands, founded the New Games Foundation in 1974, and left a lasting mark on physical education under the motto *Play Hard, Play Fair, Nobody Hurt*. It faded through the 1980s; the Foundation dissolved in 1983.

Both are prior art of the most direct kind — Orlick's book sets cooperative versions of familiar competitive games beside its new ones — and both are the reason this paper cannot honestly describe its category as new. They are also the outcome any new attempt should expect by default.

The distinction on which everything else here rests:

A cooperative game removes the opponent. A circulation sport keeps the opponent and changes what counts as a point.

This is not a rhetorical repair. A circulation sport in its adversarial mode still has sides, real opposition, a contested outcome, and a decided ending; what it changes is that the score increments on a *completed transfer* and cannot be incremented by the other side's failure. The opponent is relocated rather than removed — from a person, to the drop, to the circle's own earlier record.

COOPERATIVE GAMES	CIRCULATION SPORTS
opponent REMOVED	opponent KEPT
nobody loses	an outcome is decided
the point is inclusion	the point is the completed transfer
converts by SUBTRACTION	converts by RE-SEATING the scoring unit

If that distinction does not hold under examination, then this paper is a 1978 idea with a new name, and the correct response is to read Orlick instead. We think it holds, for a reason that is testable rather than rhetorical: **a sport can spread where a movement did not**. The cooperative-games movement asked participants to adopt a value. A circulation sport asks them to play a game whose rules happen to have that value built in, against opponents they are still trying to beat. Those are different asks with different adoption curves, and §11 pre-registers the tests.

Open question, found on revision (2026-10-01). The practitioners' own vocabulary draws this line elsewhere. USA Volleyball's *Simplified Youth Rules* describe "cooperative scoring", in which "the ball is

the opponent, and both teams work together to get as high a score as possible by counting each three contacts per side as a point" — a completed transfer as the point, with the opponent relocated to the ball exactly as the paragraph above relocates it to the drop, under the name *cooperative*. Read by that usage, relocating the opponent is itself a cooperative move, and the distinction above separates circulation from cooperation cleanly only in the adversarial mode, where two sides contest an outcome and each scores on its own completed transfers. The distinction is stated above as the authors drew it; whether it should be narrowed to the adversarial mode is left open, and §12 carries it.

11 • Pre-Registrations

Six predictions, with instruments and stated nulls — P-CS1 to P-CS5 recorded before any data, P-CS6 registered in this paper after the census and before any wider one. **The architecture forbids the obvious instrument:** the reference implementation is local-first with no telemetry by design, so every instrument below is observation, coach or teacher log, or third-party study. *The guards that make the product good make the predictions hard to test, and that is a cost being paid deliberately.*

The four originating predictions were stated by the founder on 2026-08-05 and are preserved verbatim in the project record; three of the four were unfalsifiable as stated (two vacuous conditionals, one of them also confounded by a secular trend, and one undated claim), and the re-specifications below were ratified the same day, **before any data existed**. That ordering is what makes the sharpening legitimate rather than a retrofit.

P-CS1 — Warm-up extension. In teams adopting a circulation warm-up ($n \geq 3$), median warm-up minutes per session in the final third of a season exceed the first third; and, the stronger instrument, at least one occasion on which **players ask to continue past the coach's stop**. Two further legs test the category rather than the product: **(c)** players run a circle with **no coach present**; **(d)** *who owns the clock* — does the circle play to an ending the game supplies, or to an ending someone supplies? **Null:** flat or declining minutes, no unprompted re-ask, no coach-absent play. Leg (d) requires a single session and is the cheapest instrument in this set.

P-CS2 — Scoring displacement. **(a)** In circles that have played ≥ 3 sessions, unscored or count-up sessions outnumber count-down sessions. **Null:** the adversarial mode dominates once unlocked — which is a live possibility, because competition is fun. **(b)** By **2036**, at least one mainstream sport has an officially sanctioned circulatory format played at organized level. **Null:** none has.

P-CS3 — Participation, as a property of the format. Offered both a circulatory and a competitive format of the same activity, in the same population at the same time, the circulatory format's sessions are **more gender-mixed and sustain mixed-gender play longer**. **Null:** no difference. The prediction is deliberately about what a format does and never about what any group prefers. Its plausibility rests on the gap in competitive entry being culturally contingent rather than dispositional: Gneezy, Leonard and List (*Econometrica* 77(5), 2009, 1637–1664) found that among the matrilineal Khasi the gender gap in choosing competition **reverses** relative to the patriarchal Maasai. The literature it argues against — Niederle and Vesterlund (*QJE*, 2007) — is itself contested, with meta-analytic work placing the aggregate gap at roughly a third of the original estimate (Markowsky and Beblo, *Journal of Economic Behavior & Organization* 198,

2022, 139–163). **If the gap is cultural, a format change is exactly the class of intervention that could move it.**

P-CS4 — Three generations, under scoring. Mixed-generation circles **persist when scoring is introduced** — a circle of children, adults, and elders stays mixed after graduating from the unscored mode, rather than re-sorting by age. **Null:** circles age-sort the moment the score appears. This is specified against a counterexample that already exists: **pickleball achieves three-generation play without circulation**, through a low skill floor, a slow ball, and a small court. What pickleball cannot deliver is persistence *under scoring with equal touch*, because targeting the weaker opponent is a legal and standard tactic there — which makes persistence-under-scoring a genuine discriminator rather than a restatement.

P-CS5 — The warm-up census. Run 2026-08-07. Result: partial null. Properties (i) and (iii) confirmed as predicted; **property (ii), reach, predicted rare and found common in four architectures.** The uniqueness claim is narrowed accordingly, and the narrowing is labeled post-data in §8.5. This prediction is closed. Its successor is stated below and is **not** yet run.

P-CS6 — The predictability price (new, pre-registered here, pre-data). Across a wider census including observational field data rather than published coaching materials alone, **no reach-guaranteeing format will be found whose receiver order is unpredictable across unbounded repeated play** — excluding formats using an external announcing party, which is the case under test. **Null:** such a format exists and is documented, in which case the residual contribution of the caller mechanism is smaller still and should be restated again.

The stopping rule. The full set is reviewed **every 7 January** — a date this institution already keeps for other reasons — and each prediction marked *unrun · running · resolved*. A prediction with no review date is not registered; it is merely written.

12 · Honest Limits

n = 0 for the product; n = 1 for the category's newest member. At first publication (2026-08-07), no circulation sport designed with this specification in hand had been played by anyone outside a single family, and this revision records no later trial. Every claim about adoption, enjoyment, or persistence is untested.

The census is a desk survey, not fieldwork. §8 surveyed *published* coaching materials. What coaches actually run, and how much they modify it, is exactly the sort of thing the published record misrepresents. The census's own successor prediction (P-CS6) is stated in the knowledge that field data could overturn §8.5 as readily as §8 overturned P-CS5.

One prediction in this paper has already failed. That is stated in §8 rather than buried here, but it belongs in the limits too, because it is evidence about the authors' calibration on precisely the question this paper exists to answer.

The conjecture in §8.5 about attention and predictability is unsupported. It is the most interesting output of the census and the least tested claim in the paper.

The category may be a preparation phase rather than a class. §9's falsification is real and is not rhetorical hedging. The census found nearly every mainstream member sitting inside a practice session and stopping when a coach says so.

The distinction from cooperative games carries the entire paper. If keeping the opponent while re-seating the scoring unit turns out to be a distinction without a difference in practice — if circulation sports are adopted, and abandoned, on the same curve the New Games Movement traced — then the correct conclusion is that Orlick was right, was early, and was thorough, and that this paper added a vocabulary to a settled question. And the distinction is already under strain in the vocabulary: a national governing body calls the relocated-opponent form *cooperative scoring* (§10), so the line may hold only for the adversarial mode.

The full members rest on reading a convention as a rule. §2.2 demands that skipping be impossible as a move; the three traditional games hold reach by the convention of the circle (§8.4). Read strictly, the census contains no format holding all three properties as rules, and the class as specified is ahead of any surveyed member that fully instantiates it. Read generously, an inherited convention of a ritualised game is as binding as a written rule. The paper counted the generous reading and has not argued it.

The three properties land exactly on three commitments this institution already held. Give-forward, a dignity floor, and non-accumulation were doctrine before this category was named, and a taxonomy that confirms its author's prior commitments this neatly is either a real discovery or a well-executed projection. From inside, the two are indistinguishable. **The elegance earns the experiment; it does not replace it.**

Two of the strongest members are living cultural practices with religious and courtly histories. Kemari and chinlone are described here in the flattened vocabulary of a taxonomy, which is not how their practitioners describe them, and the flattening is a real cost of the method rather than a stylistic choice.

13 · Findings Disclosed and Dedication

This paper is a study, so what it discloses is listed here as findings rather than claims, numbered so that they can be cited, contradicted, or used, each with what would break it. Findings 7 and 8 are the results of the one census (§8), and finding 8 is post-data; the others rest on argument and cited sources, and none asserts priority. All are dedicated to the public domain; **none is a patent claim, and there is nothing here to patent.**

1. **The class.** A class of sports defined by the conjunction of three properties: scoring the completed transfer rather than the termination; guaranteeing reach to every participant as a rule rather than a norm; and accumulating no individual record (§2). *Breaks if:* the conjunction admits an activity that plainly scores terminations or keeps individual records, or excludes one that holds all three as rules (the tests §6 applies to its own boundary); or if no activity is found that holds all three as rules, which §8.4 leaves open for the traditional games.
2. **The name.** *Circulation sport*, as the name of the class in finding 1, freed and never to be registered as a mark by this institution. *Breaks if:* the phrase is shown to be in established use for a different class of sports, in which case this paper's sense is a second one and should be marked as such.

3. **The conversion operator.** The three-step procedure of §5 — re-seat the scoring unit; guarantee reach; delete accumulation — with four worked conversions (basketball, volleyball, soccer, tennis) showing that it yields playable games. *Breaks if:* a sport converted by the three steps yields a game that cannot be played, or one that fails a property although every step was followed.
4. **The reverse conversion.** The operator has documented historical runs in the opposite direction: the rattan-ball circle game was turned into a net game in Siam (1933), at Seremban (1935) and, in the run the Malayan federation later recognised, in Penang (February 1945), and was standardized as *sepak takraw* at Kuala Lumpur in 1960 — which establishes the operation as real and its direction as a choice (§4). *Breaks if:* the net game is shown not to descend from the circle game, or to have kept its score on the completed transfer, in which case these were not reverse runs.
5. **The boundary.** The negative space of §6, in particular that Ultimate Frisbee is excluded because Spirit of the Game governs conduct rather than the scoring unit, and that the rondo fails on reach because it has a designated loser. *Breaks if:* Ultimate's goal, which is itself a caught pass, is read as a transfer-seated score rather than a termination, which would make Ultimate a two-of-three case rather than an exclusion; or if the standard rondo is shown to have no designated loser.
6. **The warm-up identity.** The operator's outputs already exist as the warm-ups, youth rules and coaching games of the sports they are derived from, catalogued sport by sport and read here as one family, which makes the category empirical rather than speculative (§7). *Breaks if:* the worked conversions are shown to have no close counterpart in published coaching practice, or the correspondence holds only for the four sports chosen.
7. **The census method and its result.** A coded desk survey of 22 published formats, scored on three values (instantiated, vacuous, violated), with two-person formats discounted on reach; the falsification of the authors' own prediction that reach guarantees are rare; and the four reach architectures it documents (§8). *Breaks if:* a re-coding of the same formats from the same kind of sources finds reach guarantees rare after all, or field observation shows that the published formats misdescribe what coaches run.
8. **The predictability price** (post-data, §8.5). Every reach guarantee found in the census purchases reach with predictability, and reach guaranteed in an order that no player chooses and none can anticipate, sustained across unbounded play, was not found in the census aperture on 2026-08-07. **Stated as an empirical finding only** — the underlying design argument (*equality without randomness would be a queue; the conjunction is the mechanism*) belongs to *The Sport That Says Your Name* (2026-07-23) and is cited, not re-claimed. *Breaks if:* a documented reach-guaranteeing format is found whose order is unpredictable to every participant across unbounded play, without an external announcing party. The passer-chosen all-must-touch rule found on revision (§8.5) bounds this finding without breaking it, since its order is the passers' choice.
9. **The drill/sport differentia.** *A sport owns its own ending; a drill is stopped from outside* — necessary and not sufficient within the genus of physical skilled activity, together with the derivation chain of §9.2 and the mode-ladder application of §9.3. *Breaks if:* a game generally recognised as a sport is found whose ending is left to someone's discretion rather than supplied by its rules; or an activity that ends by its own rule, and whose count is remembered, stays a drill by common usage (§9.4 reads such a case the other way).

10. **Circulation is not cooperation.** *A cooperative game removes the opponent; a circulation sport keeps the opponent and changes what counts as a point* — the specific reason this category is not the cooperative-games movement under a new name, and the claim on which the paper's predictions stand or fall (§10). *Breaks if:* circulation sports are adopted, and abandoned, on the curve the New Games Movement traced; or cooperative-games practice is shown already to hold the opponent-keeping form — a case §10 leaves open, since USA Volleyball's "cooperative scoring" relocates the opponent to the ball.
11. **The floor of three** (§2.5). Of the three properties, only the first carries a precondition that is not a design choice: below three participants a completed transfer cannot be told apart from a return, so the definition rests on one forced floor and two chosen walls. The floor belongs to Lévi-Strauss's distinction between restricted and generalized exchange, to the kula literature, and to the evolutionary work on indirect and upstream reciprocity, and is cited, not claimed; the finding is its application to this definition. *Breaks if:* a two-party activity is shown to distinguish a forward transfer from a return.
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14 • Closing

Among the oldest games in this paper is one that has been played for more than a thousand years, and an emperor's grant founded the society that preserves it. The nearest to this institution is a folk game played at dusk in Cambodian streets, which survives the way folk games do, by being played.

Between those two there is a family of games, some related by descent and some not, and beside them the warm-ups of sports coached the world over, which share one idea: **that the interesting thing about an object passing between people is the passing, and that the moment it stops is not a victory but simply the end.**

We have named that family and written down how to make more of them. Whether anyone should is a question the next decade answers, and the answer is not in our hands — which is the correct place for it, and the reason the procedure is given away rather than kept.

Authored by Thon Ly with Miss AquariusSM, the name under which this corpus discloses its AI collaboration; the underlying models are not named, and editorial control is the author's. Dedicated to the public domain under CC0 1.0. Comments, corrections, and above all counterexamples are welcome; a category is only as good as the members it correctly refuses.